Experimental Report 03

Safety-Oriented RUL Evaluation and Robust Calibration

Remaining Useful Life prediction from multivariate sensor time series

Evaluation focus	Best calibrated MAE	Calibration gain	Target cap range
safety-oriented RUL	**9.48 cycles**	**13.38 -> 12.50 MAE**	**120-130 cycles**

Safety-oriented experiment setup

- The experimental focus moved from pure regression accuracy toward safety-oriented RUL evaluation.
- Overestimation was treated as the most critical error type: predicting failure later than it will happen can create operational risk.
- Evaluation therefore included not only MAE, but also overestimation behavior, near-failure behavior, conservative calibration, and RUL target shaping.
- Capped RUL targets were tested to reduce noise at high remaining-life values; the cap was useful, but sensitive to tuning.

Core experiment results

Experiment direction	Main idea	Observed result
CatBoost comparison	Compare CatBoost against the RandomForest baseline.	Did not improve over RandomForest.
Safety analysis	Treat overestimation as a safety-critical error rather than only minimizing average error.	Evaluation expanded beyond MAE toward safety-oriented metrics.
Conservative calibration	Apply a downward prediction shift to reduce unsafe overestimation.	Aggregated RF MAE improved from 13.38 to 12.50.
Validation-only evolution	Select one strongest validation winner from evolutionary feature search.	Exposed a failure case: validation-based selection did not transfer reliably.
Robust evolutionary search	Select feature configurations by robustness rather than one validation winner.	Best official calibrated MAE reached 9.48 cycles.
Baseline/delta/health features	Normalize sensor behavior against the early engine state and add health-oriented features.	Reached MAE 12.39 and reduced overestimation.
Robust health-feature branch	Run robust search on the health-oriented feature branch.	Reached MAE 11.55 with overestimation share 0.37.

Best tested direction: robust evolutionary aggregated RandomForest with conservative calibration. The official calibrated MAE improved from 12.50 to 9.48 cycles. Capped RUL was useful, but the cap had to be tuned carefully; the most stable tested range was around 120-130 cycles, especially 125.

Calibration and target-shaping behavior

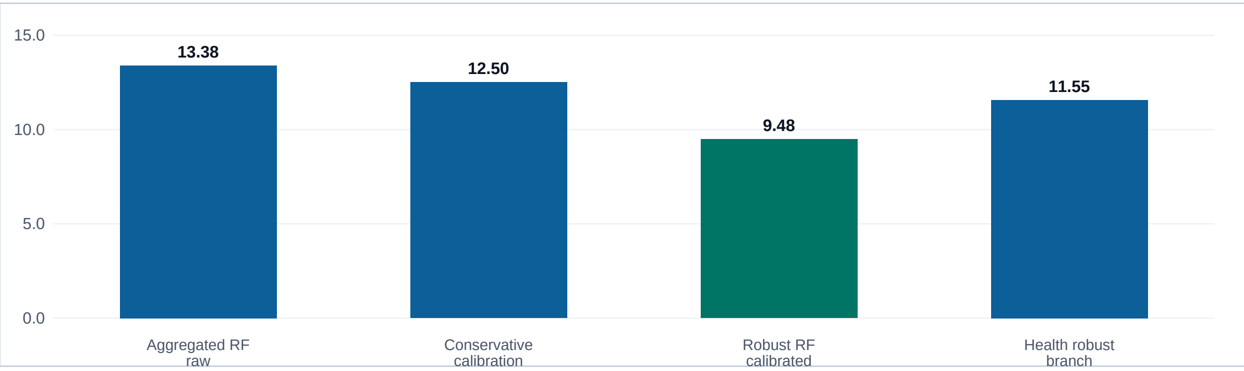

Figure 1. MAE values for the main reported evaluation stages. The strongest gain came after replacing single-winner validation selection with robust search and conservative calibration.

Domain-Aware Sensor Interpretation

Observed degradation behavior

- FD001 uses fixed operating conditions, so several sensors can remain nearly constant and carry little degradation information.
- Engine degradation does not appear as a smooth monotonic decline. Sensor traces can move in zigzags and may temporarily look healthier.
- The first tens of cycles may look close to normal because early-stage wear is weak or not visible in the sensor readings.
- An active control system can keep some sensor values near nominal conditions until late stages, making these controlled signals less useful for direct RUL prediction.
- This supports the need for filtering, windowed features, baseline normalization, and safety-aware calibration rather than relying on raw sensor trends alone.

Sensor relevance in the FD001 setting

Group	Sensors	Physical meaning / interpretation
More informative sensors	sensor_4 / column 9 sensor_9 / column 14 sensor_11 / column 16 sensor_14 / column 19	Low-pressure turbine outlet temperature; physical core speed; static pressure at high-pressure compressor outlet; corrected core speed. These are more relevant to compressor degradation and engine health changes.
Less informative in this dataset	sensor_1 / column 6 sensor_5 / column 10 sensor_10 / column 15 sensor_16 / column 21 sensor_18 / column 23	In fixed-condition FD001, some signals may be constant or actively controlled. They are not necessarily universally useless, but are less informative for this specific evaluation setting.

Safety-oriented modeling logic

Design choice	Reason for inclusion
Overestimation analysis	Predicting too much remaining life is more dangerous than predicting too little, because delayed maintenance can increase failure risk.
Near-failure evaluation	Errors close to failure are operationally more important than errors far from failure.
Conservative calibration	A downward shift in predictions can reduce unsafe overestimation while preserving useful ranking behavior.
Capped RUL target	High RUL values can be noisy and less informative; a cap around 120-130 cycles improved stability in the tested setup.
Robust feature search	Validation-only winners can fail to transfer; robust search prioritizes configurations that remain reliable under stronger checks.

Report summary

Aspect	Result
Evaluation shift	from pure MAE optimization to safety-oriented RUL evaluation
Calibration result	aggregated RandomForest MAE improved from 13.38 to 12.50
Best result	robust evolutionary aggregated RandomForest reached official calibrated MAE 9.48
Alternative feature direction	baseline/delta/health features improved interpretability and reduced overestimation
Sensor insight	fixed operating conditions and engine control make some sensors weak or misleading until late degradation